

Foundation of Blockchain Technology (IS501) Pre-Reading Material – Lecture 25

Hyper ledger Implementation of Blockchain

Estimated Reading Time: 15–20 minutes

Introduction

Hyperledger is an open-source blockchain project hosted by the **Linux Foundation**. It provides frameworks and tools for developing **enterprise blockchain applications**.

Unlike many public blockchains, Hyperledger platforms are mainly designed for **permissioned networks**, where participants are known and their access can be controlled.

Hyperledger Fabric

Hyperledger Fabric is one of the most widely used Hyperledger frameworks for enterprise blockchain implementation.

Fabric allows organizations to create a private or consortium blockchain in which:

- Participants have known identities.
- Access can be controlled.
- Transactions can be authorized.
- Organizations can maintain shared records.
- Smart contracts can implement business rules.

For example, banks, hospitals, manufacturers, and supply-chain organizations can use Fabric to share trusted information without giving complete control to a single organization.



Main Components of Hyperledger Fabric

Organizations and Peers

An organization represents a participating entity in the blockchain network. **Peers** maintain ledger data and can participate in transaction processing.

Ledger

The ledger records the transactions and the current state of the blockchain network. It provides a shared and tamper-evident record for participating organizations.

Smart Contracts

Fabric smart contracts are called **Chaincode**. Chaincode contains the business logic that defines how transactions are processed.

Certificate Authority

The **Certificate Authority (CA)** is responsible for managing digital identities and certificates for network participants.

Ordering Service

The ordering service establishes the order of transactions and distributes them to the relevant peers for validation and recording.

Channels

Channels allow a subset of network participants to communicate and maintain a separate ledger. This provides privacy between different groups within the same network.

Basic Transaction Flow

A simplified Hyperledger Fabric transaction process is:

Client Application → Endorsement → Ordering → Validation → Ledger Update

1. A client submits a transaction proposal.
2. Required peers execute the chaincode and provide endorsements.

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Course Repository: <https://github.com/rajesh101191/Foundation-of-Blockchain-Technology>



3. The transaction is sent to the ordering service.
4. Transactions are ordered and distributed to peers.
5. Peers validate the transactions.
6. Valid transactions are recorded in the ledger.

Hyperledger vs Public Blockchain

Feature	Hyperledger Fabric	Public Blockchain
Network	Permissioned	Generally permissionless
Participants	Known identities	Often anonymous/pseudonymous
Access	Controlled	Open participation
Privacy	Higher	Usually more transparent
Main Use	Enterprise applications	Public decentralized applications

Key Takeaway

Hyperledger Fabric provides a permissioned blockchain framework for implementing secure and scalable enterprise applications. Its major components include **peers, ledger, chaincode, certificate authority, ordering service, and channels.**

Think Before Class

Why might a hospital, bank, or supply-chain company prefer a permissioned Hyperledger network instead of a completely public blockchain?

